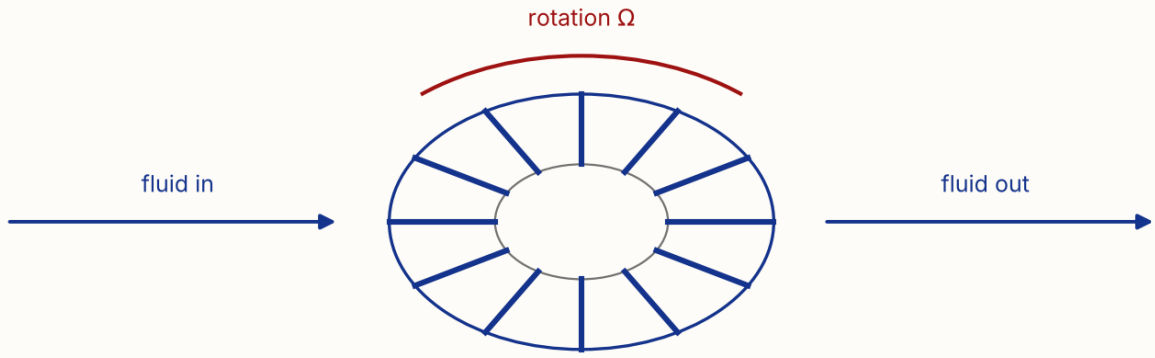
UNIT 1: Introduction and Classification

Turbo Machinery | Handwritten-style exam notes | 27 Oct 2026

Syllabus covered: Axial, radial and mixed flow machines; equations of motion in rotating frame; Coriolis and centrifugal effects; momentum and energy equation; Euler work; examples.

1. What is a turbomachine?

Rotor and stator: fluid in, fluid out, shaft work



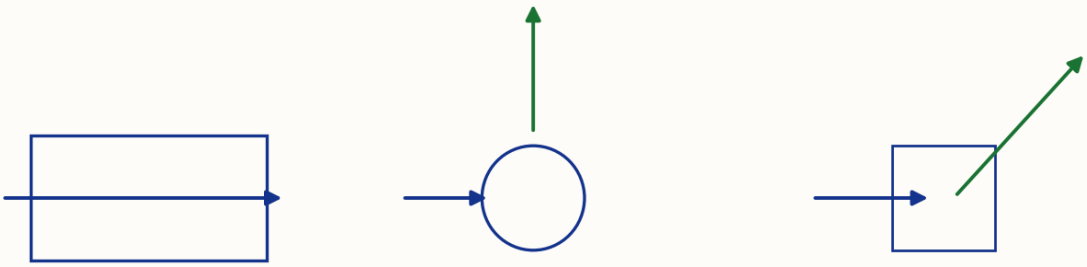
Turbine: fluid gives work to shaft. Compressor/pump: shaft gives work to fluid.

A turbomachine is a device in which **energy is transferred continuously between a rotating shaft and a flowing fluid** through a row of blades. The rotor blades change the fluid's angular momentum, and this changes its stagnation enthalpy.

- **Power producing** (fluid does work on the shaft): steam, gas and hydro turbines, wind turbines.
- **Power absorbing** (shaft does work on the fluid): pumps, fans, blowers, compressors.
- Turbomachines are preferred when large flow rates, high speed and compact size are needed.

2. Classification by flow direction

Axial, radial and mixed flow paths



The main classification is the direction of flow through the rotor:

- **Axial flow:** flow stays parallel to the shaft axis. Used in gas turbines and axial compressors. Large mass flow, moderate pressure rise per stage.

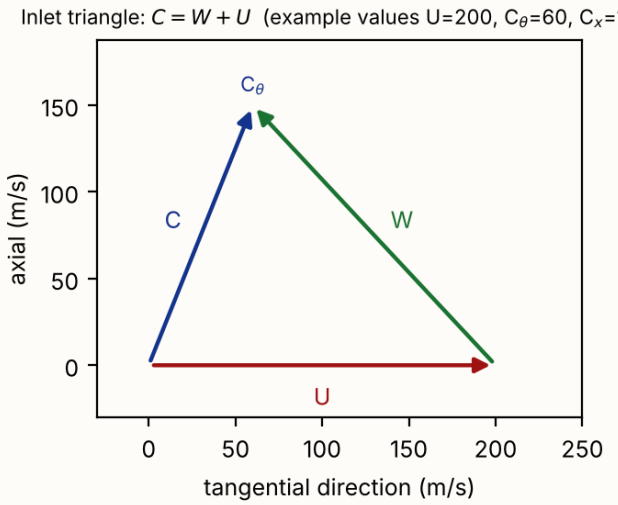
- **Radial (centrifugal) flow:** flow enters axially and leaves radially. High pressure rise per stage, wider operating range, but larger frontal area.
- **Mixed flow:** entry is axial, exit has both radial and axial components. Used in mixed-flow pumps and some turbochargers.

Feature	Axial flow	Radial flow	Mixed flow
Flow path	Parallel to axis	Radial outward/inward	Between axial and radial
Pressure rise / stage	Low to moderate	High	Moderate
Mass flow	High	Low to moderate	Moderate
Frontal area	Small	Large	Medium
Typical use	Jet engines, big fans	Small gas turbines, pumps	Mixed pumps, turbochargers

Other classifications: by energy direction (turbine vs pump/compressor), and by fluid type. Incompressible machines (pumps, hydro turbines) and compressible machines (compressors, gas and steam turbines) are analysed differently because density changes in the second group.

3. Velocity triangles (basic)

Velocity triangle: C = W + U



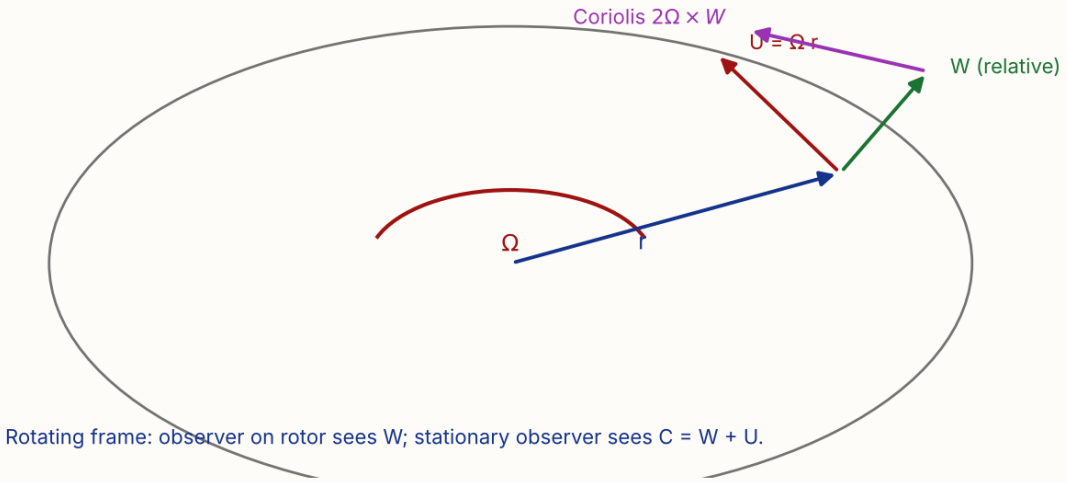
At any radius, the fluid velocity is shown as a vector triangle. Let U be the blade speed, C the absolute velocity, W the relative velocity (seen by an observer on the rotor), and alpha and beta the absolute and relative flow angles.

C = W + U where U = Ω r (C_x = axial, C_θ = tangential components)

Velocity components are used most often: C_x (axial, carries the mass flow) and C_θ (tangential, carries the angular momentum that gives work).

4. Equations of motion in a rotating frame

Rotating frame: W, U and the Coriolis direction



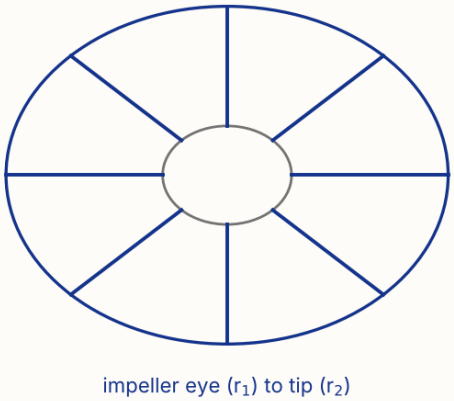
Take a rotor turning at angular velocity Ω. An observer on the rotor measures relative velocity W and relative acceleration. The absolute acceleration is related to the relative one by the extra terms of the rotating frame:

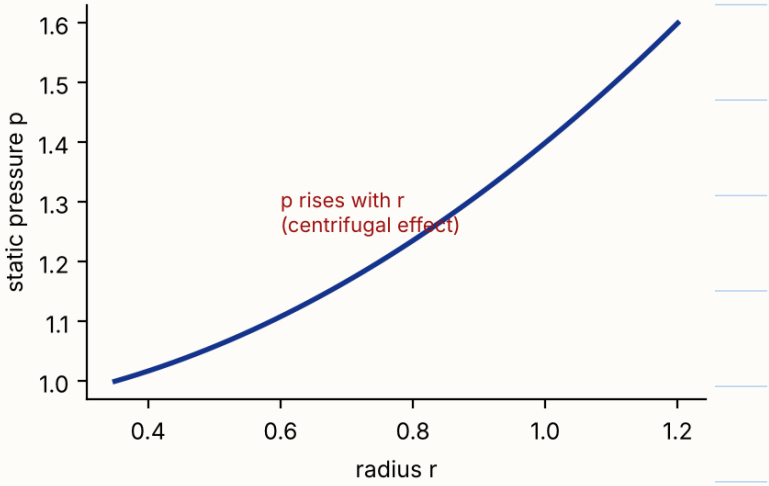
$$\mathbf{a_{abs} = a_{rel} + 2\,(\Omega \times W) + \Omega \times (\Omega \times r)}$$

- **Coriolis acceleration** $2(\Omega \times W)$: acts perpendicular to W, magnitude $2\Omega W$. It deflects the relative flow inside the rotor passage.
- **Centrifugal acceleration** $\Omega \times (\Omega \times r)$: acts radially outward, magnitude $\Omega^2 r$. It pumps fluid outward in radial machines.
- The rotating-frame momentum equation therefore has two extra source terms: $-2\rho(\Omega \times W)$ and $-\rho\Omega \times (\Omega \times r)$.

5. Effects of Coriolis and centrifugal forces

Centrifugal pressure rise in a radial impeller





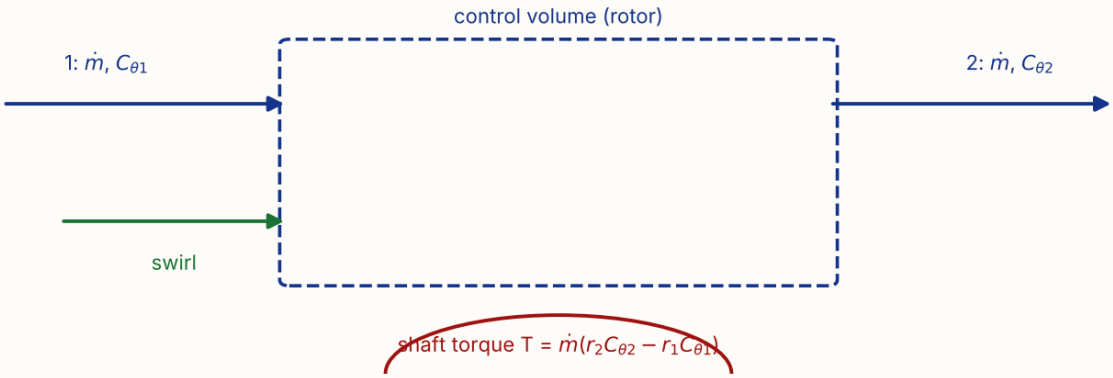
Centrifugal effect: in a radial machine the fluid is thrown outward. Even without diffusion the static pressure rises from inlet to outlet. Part of the rise comes from the rotor (centrifugal head) and part from relative velocity change:

$$p_2 - p_1 = \rho(U_2^2 - U_1^2)/2 + \rho(W_1^2 - W_2^2)/2$$

Coriolis effect: the Coriolis force acts across the relative flow, so the velocity is not uniform across the passage. Low-momentum fluid collects on one side. This non-uniform velocity causes slip (exit flow angle differs from blade angle) and can cause reversed eddies at off-design flow rates. Both are studied in Unit 4.

6. Momentum and moment of momentum equation

Control volume: torque from the change in swirl



For steady flow through a control volume, mass is conserved and the moment of momentum changes only because of torque on the fluid.

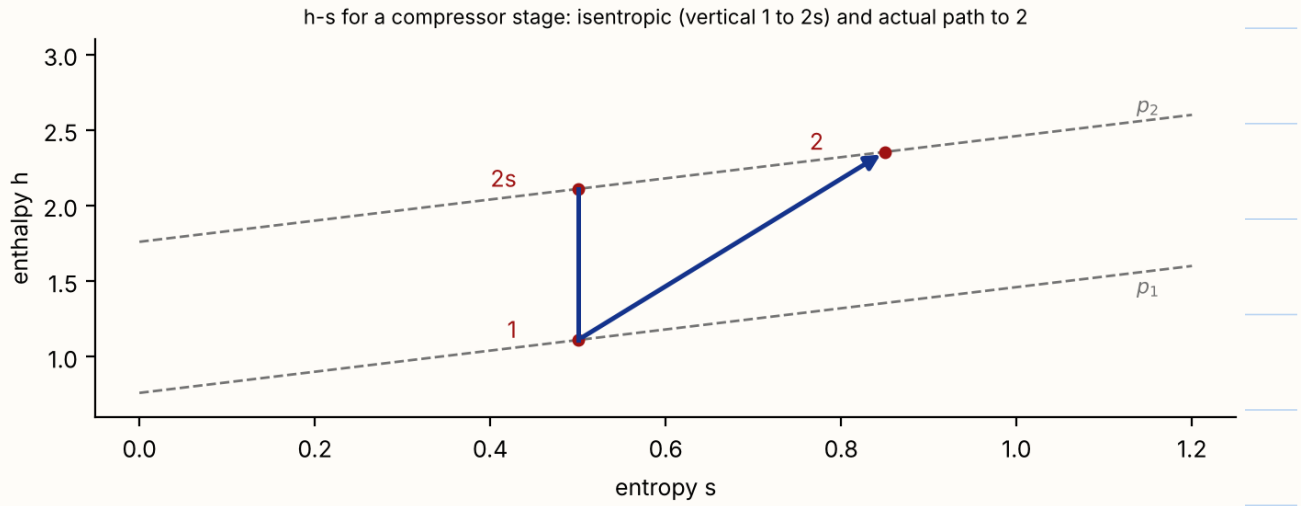
The moment-of-momentum (Euler turbomachine) equation is:

$$\mathbf{T = \dot{m}\,(r_2\,C_{\theta 2} - r_1\,C_{\theta 1}) \quad \Rightarrow \quad Power\ P = T\,\Omega = \dot{m}\,(U_2\,C_{\theta 2} - U_1\,C_{\theta 1})}$$

Here $\dot{m}$ is mass flow rate. Sign convention: C_θ is positive in the direction of blade rotation. For a compressor the power is input; for a turbine the sign of the result reverses (the fluid gives up energy).

7. Energy equation and Euler work

h-s diagram for a compressor stage



For steady, adiabatic flow with shaft work w, the energy equation is $h_{01} + w = h_{02}$. Dividing the power by mass flow gives the specific work:

$w = h_{02} - h_{01} = U_2 C_{\theta 2} - U_1 C_{\theta 1}$ (Euler work equation)

The Euler head is $H_E = w / g$ in metres of fluid. This is the ideal (no loss) head. Work done by the rotor depends only on the change in angular momentum, not on the detailed blade shape. This is why velocity triangles are the key tool.

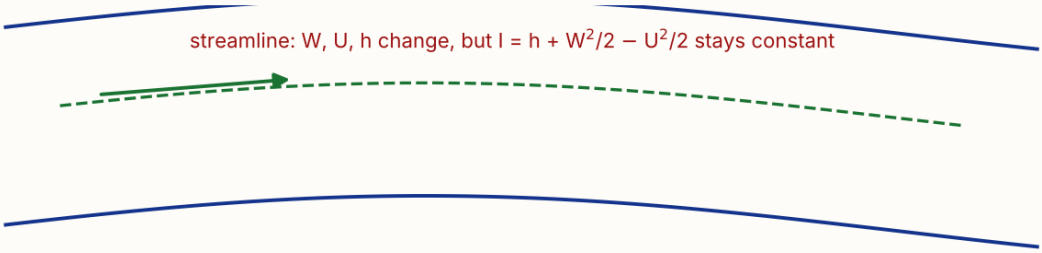
Thermodynamic link: for any element of fluid, $T\,ds = dh - dp/\rho$. In a compressor the actual work is larger than the isentropic work because of losses:

$$\eta_c = (T_{02s} - T_{01}) / (T_{02} - T_{01}) \qquad \eta_t = (T_{01} - T_{02}) / (T_{01} - T_{02s})$$

For a turbine, the actual drop is smaller than the isentropic drop, so efficiency is written with the actual drop in the numerator.

8. Rothalpy (useful in rotor analysis)

Rotor passage along a streamline



adiabatic rotor passage (schematic)

In adiabatic flow through a rotor, the rothalpy is conserved along a streamline, where it is defined as:

$$I = h + W^2/2 - U^2/2 = \text{constant along a streamline}$$

(adiabatic, no work in the relative frame)

Rothalpy shows that a change in W at constant h is linked to a change in U. It is often used to find the exit state of a rotor without solving the

whole flow field.

9. Worked examples

Working space for the examples (cover the solution and try first)



Example: *A centrifugal compressor impeller has U_2 = 350 m/s, $C_{\theta 2}$ = 250 m/s, $C_{\theta 1}$ = 0 and $\dot{m}$ = 5 kg/s. Find the specific work and the power.*

Solution: $w = U_2 C_{\theta 2} = 350 \times 250 = 87.5$ kJ/kg. Power = $\dot{m}\, w = 5 \times 87.5 = 437.5$ kW.

Example: *Air enters an axial compressor stage at T_{01} = 300 K. The stage work is 40 kJ/kg and η_c = 0.85. Find T_{02} and the isentropic work.*

Solution: c_p = 1.005 kJ/kgK, so $T_{02} - T_{01}$ = 40 / 1.005 = 39.8 K and T_{02} = 339.8 K. Isentropic work = $\eta_c \times w$ = 0.85 × 40 = 34 kJ/kg.

Example: *A pump moves 0.05 m³/s of water with a head of 30 m. Find the water power.*

Solution: P = ρ g Q H = 1000 × 9.81 × 0.05 × 30 = 14.7 kW.

10. Exam quick points

- Axial = along axis, radial = outward, mixed = both.
- Euler work: $w = U_2C_{\theta 2} - U_1C_{\theta 1}$. Remember it is the change in angular momentum.
- Coriolis causes non-uniform relative flow and slip. Centrifugal force causes static pressure rise in radial machines.
- Always state the sign convention for swirl (C_θ is positive along rotation).

11. Practice questions

Q1. Classify a jet engine compressor and a hydro-turbine by flow direction. Give one reason for each choice.

Q2. Derive the Euler work equation from the moment-of-momentum equation.

Q3. A pump has U_2 = 25 m/s and $C_{\theta 2}$ = 18 m/s with no pre-whirl. Find the Euler head (g = 9.81 m/s²).